Name :- Santosh Bhandari Subject :- Discrete Structures

Group - 'A'

Q.N-1

An algebric structures is a non-empty set equipped with one or more binary operations defined on it that satisfy specific algebric axioms. It is denoted by S, *

Soln

Let $a, b, \in G$, and $a*b = \frac{ab}{2}$, $(G, *)$ is abelian

i) Closure property

$$\frac{a*b}{2} \neq 0 \text{ as } a \neq b, b \neq 0 \text{ \& } a*b \in G$$

ii) Assocative property

$$(a*b)*c = \frac{ab}{2} * c = \frac{\frac{abc}{2} * c}{2} = \frac{abc}{4}$$

$$a*(b*c) = a * \frac{bc}{2} = \frac{a * \frac{bc}{2}}{2} = \frac{abc}{4} \quad \because a,b,c \in G$$

$\therefore (a*b)*c = a*(b*c)$ is assocative property.

iii) Identity element

$$\frac{a.2}{2} = a = \frac{2.a}{2}$$

$\therefore$ 2 is identity element.

iv) Inverse element

$$a * a^{-1} = 2 = a^{-1} * a \quad a^{-1}, a \in G.$$

$\therefore a^{-1}$ is inverse element

v) Commutative property

$$a*b = \frac{ab}{2} \text{ \& } b*a = \frac{ba}{2} = \frac{ab}{2}$$

$\therefore$ It is commutative property.

$\therefore (G, *)$ holds all the properties. It is an Abelian group.

Q.No.2. A boolean algebra is an algebric structure $(B, +, \cdot, ', 0, 1)$ consisting of ~~a~~ set B, two binary operation '+' and '·', a unary operation ''' and two special elements '0' and 1, Satisfying closure, Commutative, associative, distributive, identity and complement laws.

SOP :- It stands for Sum of product. ~~terms~~ An expression composed of product terms combined together by addition.

$$F(A,B,C) = AB + BC + A'C$$

POS :- It is product of sum which is composed of sum term combined together by multiplication.

$$F(A,B,C) = (A+B+C)\cdot(A'+B)\cdot(D+C')$$

A Boolean expression is in canonical form which every product terms contains all variables of the function in either complemented or uncomplemented form or every sum term contains all variables.

Ex:- $F(A,B) = A'B + AB' + AB$

$F(A,B) = (A'+B)\cdot(A+B')\cdot(A+B)$

Q.No.4. Let $d = \gcd(a,b)$, d/a & d/b

$\therefore$ There exists co-prime integers x, y, where ~~x,y~~ $\gcd(x,y) = 1$ such that

$a = dx$ & $b = dy$

Now, $a \cdot b = (dx)\cdot(dy) = d(dxy)$

We know that the LCM of $a = dx$ & $b = dy$ where $\gcd(x,y) = 1$ is given by

$LCM(a,b) = dxy$

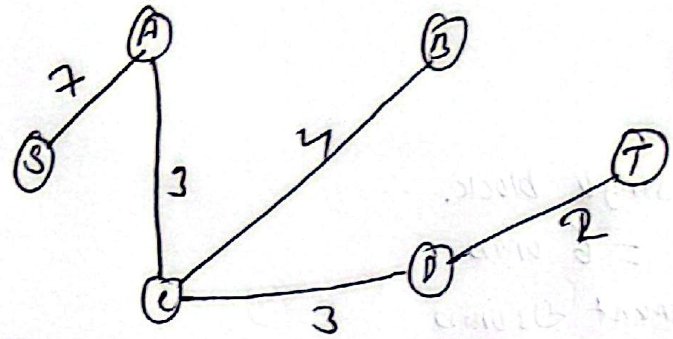
Now, $= \gcd(a,b) \cdot LCM(a,b)$

$= d \cdot (dxy) = (dx)\cdot(dy) = a \cdot b$

$\therefore \gcd(a,b) \cdot LCM(a,b) = a \cdot b$ proved.

Q.5. Solⁿ

Minimum spanning tree using Kruskal's Algorithm



Now,
The MST edges DT, CD, AC, BC & AS.

Weight = 2+3+3+4+7 = 19

Q.No. 6.

Solⁿ

Let take the vertex 'f'. The degree of vertex f in 'X' is 3.
We can map the vertex 'f' to W, Z, V, and S in graph Y.
The adjacency vertices and their degree of vertices 'f' are
b=3, e=2, g=2

Let map in W
S=3, Z=3, V=3

Map in S :- t=2, W=3, V=3
map in Z = W=3, Y=2, V=3
map in V = Z=3, V=2, S=3

∴ The degree of adjacent vertex that we are going to map doesn't match. So, it is not Isomorphic graph.

Q.no.7

Soln

Word = 'COMPUTER'

$n = 8$

vowels = $\{o, u, e\}$

a) vowels become single block.

Total arrangment = 6 units

3 vowels arrangment = 3 units

$\therefore$ Total ways $= 6! \times 3!$

$= 720 \times 6 = 4320$ ways

b) Ways to arrange 3 vowels into 4 odd positions

$$P(4,3) = \frac{4!}{(4-3)!} = 24$$

Ways to arrange 5 consonants in 5 vacant position

$$P(5,5) = 5! = 120$$

$\therefore$ Total ways $= 24 \times 120 = 2880$ ways.

Q.no-8.

~~Soln~~

A permutation that can be expressed as product of an even number of a ~~cycles~~ transportation is even permutation.

If a permutation can be expressed as an product of an odd numbers of transportation, then it is odd permutation.

Soln

$$P = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 5 & 4 & 3 & 6 & 1 & 7 & 9 \end{pmatrix}$$

cycle form $= (1,2,5,6), (3,4), (7), (8,9)$

Now,

cyclic form for Transportation $= (1,2,5,6),(3,4),(8,9)$

$= (1,2),(1,5),(1,6),(3,4),(8,9)$

The No. of transportation is odd. So the permutation is also odd.

Q.No. 4.

Soln

R is equivalance if an only if

i) Reflexive :-

$$x - x = 0 = 0.m \quad \forall x \in Z$$

$\therefore (x,x) \in R$, So it is reflexive

ii) Symmetric :-

$(x,y) \in R \Rightarrow x \equiv y \pmod m$

$x - y = k.m \quad \forall k \in Z$

$y - x = -k.m = (-k).m, \quad -k \in Z$

$\therefore y \equiv x \pmod m \Rightarrow (y,x) \in R$ $\therefore$ It is symmetric

iii) Transitive :-

$(x,y) \in R$ & $(y,z) \in R$

$x - y = k_1 m$ & $y - z = k_2.m, \quad k_1, k_2 \in Z$

$(x-y) + (y-z) = k_1 m + k_2 m = (k_1 + k_2) m$

$\therefore x - z = (k_1 + k_2) m$

$\therefore k_1 + k_2 \in Z, \; x \equiv z \pmod m \Rightarrow (x,z) \in R$

$\therefore$ It is transitive

$\therefore$ The R is reflexive, symmetric and transitive, so it is equivalance relation.

Q.No. 10. An argument is a sequence of statement followed by a final statement i.e. conclusion. An argument is valid if the truth of all premises logically guarantees the truth of the conclusion.

Now,

p = It has rained, q = The ground is wet.

In symbolic form :-

$P_1 = p \rightarrow q$

$P_2 = p$

$C = q$

Truth Table

p	q	p→q
T	T	T
T	F	F
F	T	T
F	F	T

In Row 1, both the premises are true and the conclusion is also true. Thus the given statement/argument is valid.